KAIZEN MATHS

Lesson starter

Upper and Lower Bounds

Calculations With Bounds

Name

S1

Find the required bound for each calculation.

a = 39 m to the nearest 0.1 and b = 16.9 m to the nearest 0.1. Find the upper bound of a + b.

S2

Find the required bound for each calculation.

d = 175 m to the nearest whole number and t = 55.8 s to the nearest 0.1. Find the lower bound of d t .

S3

Find the required bound for each calculation.

a = 27 m to the nearest whole number and b = 28 m to the nearest whole number. Find the upper bound of a + b.

S4

Find the required bound for each calculation.

d = 155 m to the nearest whole number and t = 48.8 s to the nearest 0.1. Find the lower bound of d t .

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Calculations With Bounds

Name

EA1

Find the required bound for each calculation.

a = 24 m to the nearest whole number and b = 20.1 m to the nearest 0.1. Find the upper bound of a - b.

EA2

Find the required bound for each calculation.

d = 227 m to the nearest whole number and t = 39.3 s to the nearest 0.1. Find the upper bound of d t .

KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Calculations With Bounds

Name

EB1

Find the required bound for each calculation.

a = 11 m to the nearest whole number and b = 10 m to the nearest whole number. Find the lower bound of a - b.

KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Calculations With Bounds

Name

EA1

Find the required bound for each calculation.

a = 24 m to the nearest whole number and b = 20.1 m to the nearest 0.1. Find the upper bound of a - b.

EA2

Find the required bound for each calculation.

d = 227 m to the nearest whole number and t = 39.3 s to the nearest 0.1. Find the upper bound of d t .

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Show your working

KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Calculations With Bounds

Name

EB1

Find the required bound for each calculation.

a = 11 m to the nearest whole number and b = 10 m to the nearest whole number. Find the lower bound of a - b.

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Show your working

Teacher answers & working

Upper and Lower Bounds · Calculations With Bounds

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S1

Find the required bound for each calculation.

a = 39 m to the nearest 0.1 and b = 16.9 m to the nearest 0.1. Find the upper bound of a + b.

Answer:

56 m

Working:

Write the error interval for a: 38.95 m ≤ a < 39.05 m.

Write the error interval for b: 16.85 m ≤ b < 16.95 m.

Use the upper bound of both values.

Upper bound of a + b = 39.05 + 16.95 = 56.

Therefore the required bound is 56 m.

S2

Find the required bound for each calculation.

d = 175 m to the nearest whole number and t = 55.8 s to the nearest 0.1. Find the lower bound of d t .

Answer:

3.124 m/s

Working:

Write the error interval for d: 174.5 m ≤ d < 175.5 m.

Write the error interval for t: 55.75 s ≤ t < 55.85 s.

To make d/t as small as possible, use the lower bound of d and the upper bound of t.

Lower bound = $\frac{174.5}{55.85}$ = 3.124.

Therefore the required bound is 3.124 m/s.

S3

Find the required bound for each calculation.

a = 27 m to the nearest whole number and b = 28 m to the nearest whole number. Find the upper bound of a + b.

Answer:

56 m

Working:

Write the error interval for a: 26.5 m ≤ a < 27.5 m.

Write the error interval for b: 27.5 m ≤ b < 28.5 m.

Use the upper bound of both values.

Upper bound of a + b = 27.5 + 28.5 = 56.

Therefore the required bound is 56 m.

Teacher answers & working

Upper and Lower Bounds · Calculations With Bounds

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S4

Find the required bound for each calculation.

d = 155 m to the nearest whole number and t = 48.8 s to the nearest 0.1. Find the lower bound of d t .

Answer:

3.163 m/s

Working:

Write the error interval for d: 154.5 m ≤ d < 155.5 m.

Write the error interval for t: 48.75 s ≤ t < 48.85 s.

To make d/t as small as possible, use the lower bound of d and the upper bound of t.

Lower bound = $\frac{154.5}{48.85}$ = 3.163.

Therefore the required bound is 3.163 m/s.

EA1

Find the required bound for each calculation.

a = 24 m to the nearest whole number and b = 20.1 m to the nearest 0.1. Find the upper bound of a - b.

Answer:

4.45 m

Working:

Write the error interval for a: 23.5 m ≤ a < 24.5 m.

Write the error interval for b: 20.05 m ≤ b < 20.15 m.

Use the upper bound of a and the lower bound of b.

Upper bound of a - b = 24.5 - 20.05 = 4.45.

Therefore the required bound is 4.45 m.

EA2

Find the required bound for each calculation.

d = 227 m to the nearest whole number and t = 39.3 s to the nearest 0.1. Find the upper bound of d t .

Answer:

5.796 m/s

Working:

Write the error interval for d: 226.5 m ≤ d < 227.5 m.

Write the error interval for t: 39.25 s ≤ t < 39.35 s.

To make d/t as large as possible, use the upper bound of d and the lower bound of t.

Upper bound = $\frac{227.5}{39.25}$ = 5.796.

Therefore the required bound is 5.796 m/s.

Teacher answers & working

Upper and Lower Bounds · Calculations With Bounds

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EB1

Find the required bound for each calculation.

a = 11 m to the nearest whole number and b = 10 m to the nearest whole number. Find the lower bound of a - b.

Answer:

0 m

Working:

Write the error interval for a: 10.5 m ≤ a < 11.5 m.

Write the error interval for b: 9.5 m ≤ b < 10.5 m.

Use the lower bound of a and the upper bound of b.

Lower bound of a - b = 10.5 - 10.5 = 0.

Therefore the required bound is 0 m.